

Lab 17: Neural Networks

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17.1 Play Action or Run?

Use `17_play-action-vs-run.csv` to study a football question before the snap: when a play is either a designed run or a play action pass, can the alignment at the snap distinguish one from the other?

Each row is one 2022 NFL play frozen at the snap. At that moment, the offense has reached its final alignment and the defense has declared its final structure. All coordinates have been standardized so the offense moves to the right.

Ordinary dropbacks without play action have been removed. The remaining response is

$$Y_i = \begin{cases} 1, & \text{play action dropback pass,} \\ 0, & \text{designed run.} \end{cases}$$

This is not a generic pass versus run problem. The positive class is a pass designed to look like a run, and the negative class is an actual run. The goal is to evaluate how much play action deception is visible before the ball is snapped.

The features visible before the snap include:

- game context: quarter, down, distance, field position, clock, score differential, and expected points;
- formation context: offensive formation, receiver alignment, and whether there was motion or a shift before the snap;
- alignment geometry: offensive and defensive width, depth, speed, box count, and nearest defender summaries.

Use the prepared split by week:

- train on Weeks 1–6: 4,189 runs and 1,723 play action passes;
- validate on Week 7: 644 runs and 230 play action passes;
- test on Weeks 8–9: 1,279 runs and 516 play action passes.

Use Weeks 1–6 to fit candidate models and Week 7 to choose tuning parameters. After the modeling choices are fixed, refit the selected models on Weeks 1–7 when feasible and report final test log loss on Weeks 8–9. Do not use the test weeks to choose features, tuning parameters, architectures, or plotting choices.

17.1.1 Task 1: Baselines and Formation Signal

Load the data and report the class balance in each split. Then fit two baselines:

1. **Training mean:** predict the training set play action rate for every play.
2. **Formation rate:** predict the training set play action rate within each formation and alignment cell. For cells not seen in training, use the training mean.

Report validation and test log loss for both baselines. Then, using only the training data, make a plot or table of play action rate by formation and receiver alignment. Which cells have the highest play action rates? Which have the lowest?

17.1.2 Task 2: Logistic Regression and Feedforward Neural Network

Fit a logistic regression model using the features visible before the snap. This is the linear baseline for the tabular feature set.

Next, fit a feedforward neural network with one hidden layer using the same inputs. Use validation log loss to tune

$$H \in \{2, 4, 6, 8\}$$

hidden units and a grid of weight decay penalties. For each candidate, train on Weeks 1–6 and evaluate on Week 7. Report the validation grid, the selected number of hidden units, and the selected weight decay value.

17.1.3 Task 3: XGBoost Benchmark and Final Comparison

Fit an XGBoost model as a nonlinear tabular benchmark. Tune at least:

- maximum tree depth;
- learning rate;
- minimum child weight;
- number of boosting rounds, using early stopping on validation log loss.

After selecting each model's tuning parameters, evaluate all final models on the test weeks. Create a table with one row per model:

Model	Selected tuning parameters	Validation log loss	Test log loss
Training mean			
Formation rate			
Logistic regression			
Feedforward neural network			
XGBoost			

Then make a bar plot of final test log loss values. Which model predicts play action passes best? Explain why that model's structure may suit this problem well. If XGBoost performs best, your explanation should discuss how trees split on thresholds and build interactions among formation, down, distance, spacing, and box count.

17.1.4 Task 4: From Snap Frames to CNN Channels

Use `17_snap-examples.csv` to inspect two example plays: one play action pass and one run. The file contains coordinates at the snap for the 22 players and the football.

For one example play:

1. Create an offensive channel and a defensive channel on a coarse field grid.
2. Plot the two channels side by side.
3. Describe what information is preserved by this representation and what information is lost compared with the tabular feature row.

You do not need to train a convolutional neural network in this lab. Instead, explain what a 3×3 filter would see in your grid. What choices define the CNN before training? Which quantities would the model learn from data?

References

- [Rumelhart1986] Rumelhart, D. E., Hinton, G. E., & Williams, R. J., *Learning Representations by Back-Propagating Errors*, Nature, Vol. 323, pp. 533–536, 1986.
- [ChenGuestrin2016] Chen, T., & Guestrin, C., *XGBoost: A Scalable Tree Boosting System*, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.